Phishing Email Detection Using Machine Learning and Deep Learning

Team Members

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Course

CSC 490 – Senior Project

Instructors

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Abstract

Phishing remains one of the most persistent and evolving threats in cybersecurity, relying on social engineering techniques to deceive users into disclosing sensitive information. As phishing emails become increasingly sophisticated, traditional rule based detection systems struggle to adapt to new attack patterns. This project explores the use of machine learning and deep learning techniques to automatically detect phishing emails with high accuracy.

To address this problem, multiple publicly available datasets including CEAS, Enron, Ling, and Nigerian phishing datasets—were combined to create a diverse and representative dataset. A significant portion of the project involved cleaning and standardizing these datasets, as inconsistencies in formatting and labeling initially led to unreliable model performance. Through iterative preprocessing and validation, the dataset was refined to support effective training.

Three models were implemented and evaluated: Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and a Convolutional Neural Network (CNN). The results showed that while traditional models such as SVM performed strongly, the CNN model achieved the highest

overall performance, demonstrating the advantages of deep learning in capturing complex patterns in text data.

This project highlights the effectiveness of machine learning in cybersecurity applications and provides insight into the importance of data quality, model selection, and iterative development in building reliable detection systems.

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1. Introduction

Phishing attacks continue to be a major concern in modern cybersecurity, targeting individuals and organizations through deceptive and carefully crafted emails. These emails often mimic legitimate sources, making them difficult to detect using traditional filtering techniques. As attackers refine their methods, it becomes increasingly important to develop intelligent systems capable of identifying phishing attempts based on patterns and behavior rather than static rules.

The motivation behind this project was to explore how machine learning and deep learning techniques can be applied to improve phishing detection. Rather than relying on predefined rules, this approach allows models to learn patterns from data and adapt to new forms of attacks. This project aims to build and evaluate such a system while gaining practical experience with real-world datasets and machine learning workflows.

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2. Project Description

The objective of this project was to design and implement a system capable of classifying emails as either legitimate or phishing. This involved addressing several challenges, including handling multiple datasets with different structures, extracting meaningful features from text, and selecting appropriate models for classification.

The project followed a structured workflow:

- Collect and combine multiple phishing datasets
- Perform data cleaning and preprocessing
- Extract features from email text
- Train and evaluate multiple machine learning models
- Compare performance and analyze results

Throughout development, it became clear that preprocessing played a critical role in overall system performance. Early experiments showed that inconsistent data significantly reduced model accuracy, reinforcing the importance of thorough data preparation.

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3. Software and Hardware Specifications

Software

- Python
- Jupyter Notebook
- Scikit-learn
- TensorFlow / Keras
- Pandas and NumPy
- Matplotlib

Hardware

- Personal computer
- Optional GPU support for faster deep learning training

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4. Technical Details

4.1 Data Preprocessing

The datasets used in this project varied significantly in structure and content, requiring extensive preprocessing. Initial attempts to train models on raw data resulted in inconsistent and unreliable performance. To address this, the following steps were applied:

- Removal of noise such as special characters and irrelevant symbols
- Conversion of text to lowercase for consistency
- Tokenization of email content
- Handling missing or incomplete data
- Standardization of labels across datasets

This iterative cleaning process significantly improved model performance and stability.

4.2 Feature Engineering

Feature engineering focused on transforming raw email text into a format suitable for machine learning models. Techniques such as TF-IDF vectorization were used to represent text numerically.

Due to team restructuring, feature engineering responsibilities were shared between Khalil Simmons and Yuuki Moody. This required coordination to ensure consistency across datasets and model inputs.

4.3 Model Implementation

Support Vector Machine (SVM)

SVM was selected due to its effectiveness in high-dimensional spaces, particularly for text classification problems. It provided strong baseline performance and demonstrated reliability across different datasets.

K-Nearest Neighbors (KNN)

KNN was implemented as a simpler, distance-based model. While easy to understand and apply, it proved less effective for this problem due to its sensitivity to data distribution and inability to generalize well with high-dimensional text data.

Convolutional Neural Network (CNN)

The CNN model was used to capture deeper patterns within the text. Unlike traditional models, CNNs can learn hierarchical features, making them well-suited for identifying subtle characteristics of phishing emails. This model required more tuning and computational effort but ultimately produced the best results.

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5. Test Results

The models were evaluated using accuracy, precision, recall, and F1-score.

Performance Overview

- CNN achieved approximately 99% across all metrics

- SVM achieved strong performance around 97–98%
- KNN performed significantly lower, with results ranging from 63–80%

Analysis

The results demonstrate that model performance is heavily influenced by both data quality and model complexity. The CNN model outperformed others due to its ability to learn complex relationships within the text, which are often missed by simpler models.

SVM also performed well, indicating that traditional machine learning methods remain effective when combined with proper feature engineering. In contrast, KNN struggled due to its reliance on distance metrics, which are less effective in high-dimensional feature spaces.

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6. Conclusion

This project successfully developed a phishing email detection system using both machine learning and deep learning techniques. Through iterative preprocessing, model training, and evaluation, the system achieved high accuracy and demonstrated the effectiveness of data-driven approaches in cybersecurity.

The results highlight the importance of selecting appropriate models and investing time in data preparation. The CNN model emerged as the most effective approach, reinforcing the value of deep learning in modern classification tasks.

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7. Future Work

Future improvements could focus on:

- Expanding the dataset to improve generalization
- Implementing real-time detection capabilities
- Exploring advanced models such as LSTM or transformer-based architectures
- Deploying the system as a user-facing application

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8. Reflection

This project provided valuable experience in applying machine learning techniques to a real-world cybersecurity problem. One of the most important lessons learned was the impact of data quality on model performance. Early challenges with inconsistent datasets highlighted the need for careful preprocessing and validation.

Additionally, the project required adaptability in response to team changes. Responsibilities were adjusted to ensure all components were completed, demonstrating the importance of collaboration and communication in group projects.

Overall, this experience strengthened both technical and problem-solving skills, as well as the ability to manage a project from initial concept to final implementation.

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9. References

- Scikit-learn Documentation
- TensorFlow Documentation
- CEAS Phishing Dataset
- Enron Email Dataset
- Research on phishing detection using machine learning

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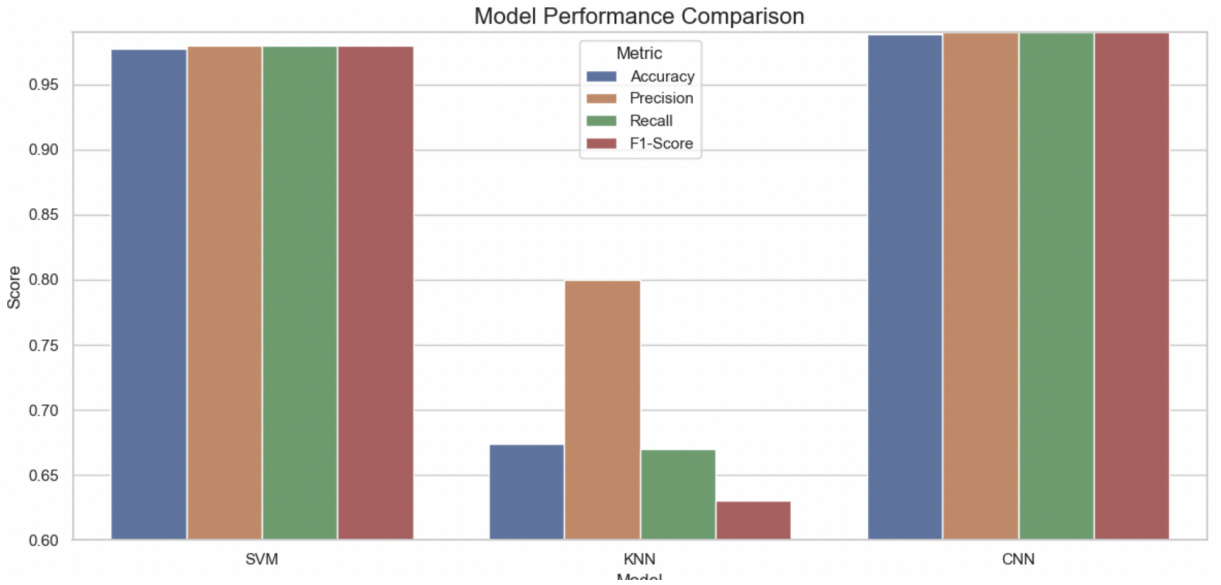
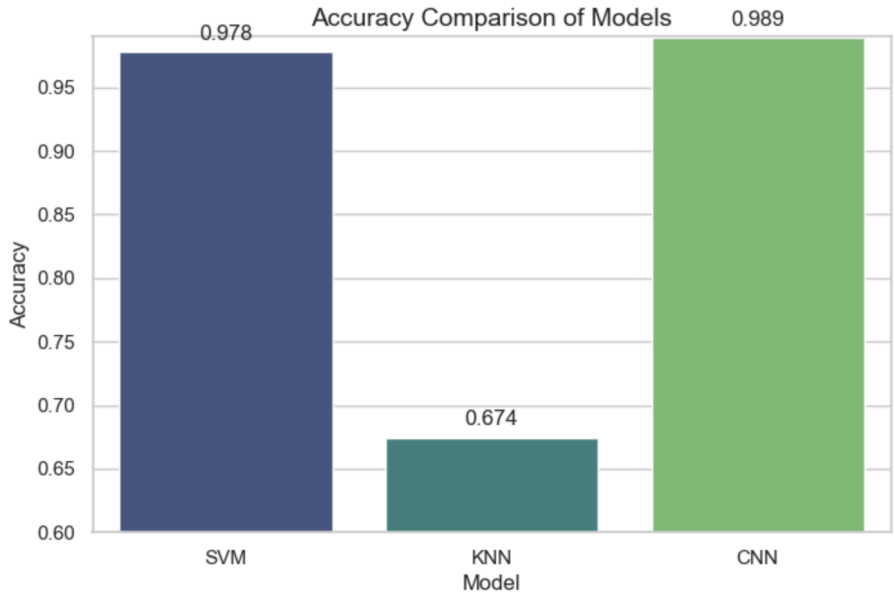
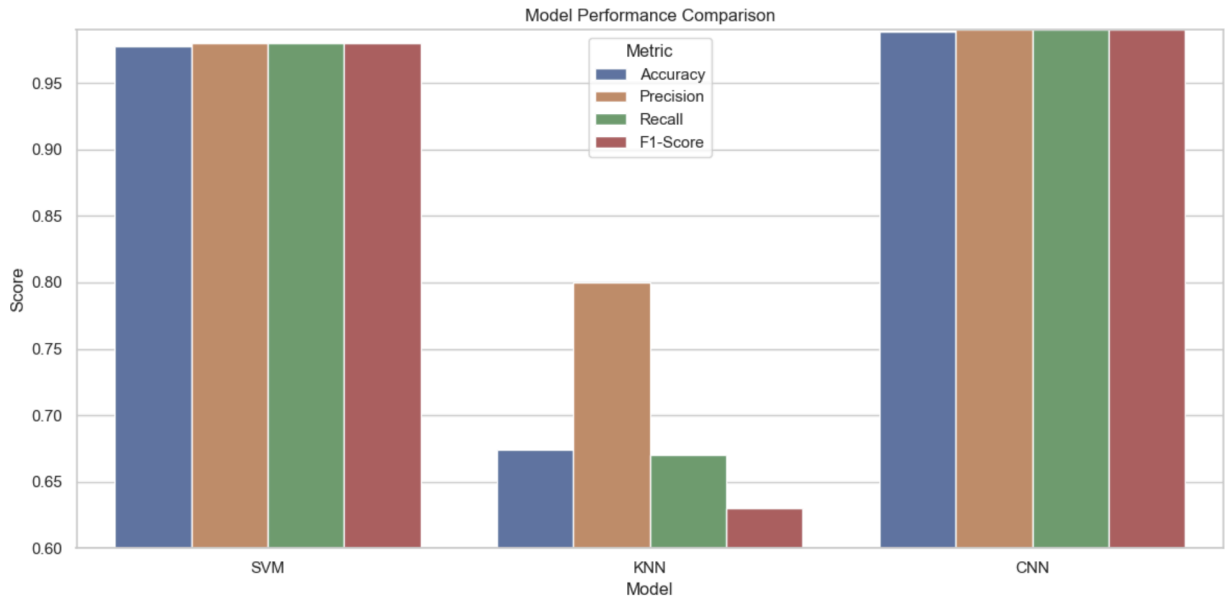
10. Work Log

Team Member	Contribution	Percentage
Khalil Simmons	Project coordination, feature engineering, workflow management	30%
Hamid Hassan	Dataset preprocessing and preparation	25%
Yuuki Moody	Model development and CNN implementation	25%
Joseph Woodard	Documentation, reporting, and presentation	20%

GitHub Repository: https://github.com/khalilLS611/New-phishing-email-detection/tree/main


Model Performance Comparison
Metric
Accuracy
Precision
Recall
F1-Score
Score
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
SVM
KNN
CNN


Accuracy Comparison of Models
0.978
0.674
0.989
Accuracy
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
SVM
KNN
CNN
Model


Model Performance Comparison
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